

a security key online is US\$ 18. Hence, it can be a fairly inexpensive option for an easy-to-use additional layer of security, especially for online financial transactions.

OpenID – distributed authentication

While OpenID does not eliminate the need for a password, it does reduce the number of passwords you may need. According to Wikipedia:

As of March 2016, there are over 1 billion OpenID enabled accounts on the Internet and approximately 1,100,934 sites have integrated OpenID consumer support.

If the website providing the identity services switches to an authentication process that doesn't rely exclusively on passwords, every site using its services benefits.

The drawback of OpenID is one of trust.

The identity provider does, however, get a log of your OpenID logins; it knows when you logged into what website, making cross-site tracking much easier. A compromised OpenID account is also likely to be a more serious breach of privacy than a compromised account on a single site. (en.wikipedia.org/wiki/OpenID)

The perceived drawback is an incentive for serious OpenID providers to ensure greater security for authentication and win the trust of users.

A plan for traveller identification

When travelling abroad, people from various organisations need to verify your details like your passport, visa, tickets, etc. This takes time, as can be seen with the long lines at airline and immigration counters. If you lose or misplace any of the documents required, it becomes a very serious issue. And losing/misplacing documents during the fraught, tension-filled environment of international travel is not all that difficult.

Sita, an organisation in the travel trade, hopes to address the issue by encrypting the travel documents using public key encryption. The encrypted data would be stored on a distributed database (which was created for bitcoins) using block chains. Travellers would give their public key (quite possibly using a 2D bar code) during the journey, where ever needed, for a faster and simpler travel experience.

This plan and the security key are examples of higher security with greater ease of use. A common thread in all the above-mentioned cases is the use of private/public key encryption.

Using OpenPGP in Firefox

Accessing OpenPGP keys in a browser needs a plugin. Your Google search will lead you to WebPG and Firepgg. The latter project has been discontinued. WebPG, which seems to work for some and not for others, was last updated in January 2013. So, here is an opportunity for some students to revive or create a project to enhance the security of Web applications, e.g., by enabling users to seamlessly and digitally sign their critical data.

As an increasing amount of our data and transactions are online, with fewer human interfaces to resolve unexpected failures, it is imperative that the developers think of security first and get rid of the meaningless inanities, like a typical line at the end of various statements, which states: "This is a computer generated document and needs no signature."

The user should be able to trust the site and, more significantly, the site should be able to identify and trust the user! 

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